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Non-Destructive Evaluation of Oil Palm Fresh Fruit Bunch Quality Using Thermal Vision

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Abstract. Oil palm (*Elaeis Guineensis* Jacq) is the main product of Indonesian agro-industry. It supports the country's economy. After harvest, the Fresh Fruits Bunch (FFB) should immediately be processed to extract the oil. Delayed will reduce the oil quality significantly. Nonetheless, due to poor road-conditions, the FFB often delivered to the palm mills beyond the acceptable time. To reduce the decrease of the fruit freshness, generally cold storage was applied. In this study, the FFB immediately freeze after harvest. Fruits stored at -20 °C, upon delivered to the mills. On arrival, acclimatization process was perform to restore fruit temperature before processing. A pseudo-color thermal vision used to monitor the fruits surface temperature continuously. Various fruits maturities observed in this study were: 120, 140, 160, 180, and 200 days after antheses (DAA). Temperature calibration was done by manually measured the fruit's surface using a thermometer. Models developed to correlate the actual surface temperature of fruits with pseudo-color displayed by the thermal-vision. Results showed the model of thermal vision correctly measured fruits temperature with R^2 of 0.979, 0.973, 0.976, 0.976, and, 0.973 for stored fruits harvested at 120, 140, 160, 180, and 200 DAA respectively. The color of images on the camera can be used to observe the temperature of the fruits. The highest correlation obtained with redness index (R^2 of 0.7677-0.8037), and the lowest is bluish index (R^2 of 0.1101-0.561). FFB harvested at 180 DAA has the strongest correlation (R^2) of temperature when observed by thermal-vision.

1. Introduction

Palm oil is the main plantation commodity in Indonesia, and support the economy [1]. The vegetable oil derived by milling the fruits. It mainly consumed by industrial sector, including food oleochemicals and industry [2]. The oil quality is strongly influenced by the postharvest handling and fruits ripeness upon harvest. Properly harvest and handle the Fresh Fruits Bunch (FFB) can prevent the damages [3].

The FFB should immediately process upon harvest to maintain the freshness. Milling the fruit in fresh condition (less than 22 hours after harvesting) will prevents the deterioration of oil due to enzymatic activities and oxidation process [4]. Delayed will increase the level of free fatty acids (FFA) in oil [5]. The FFA formed due to the oxidation and enzymatic process. The FFA level indicates the oil quality. A high FFA level showed a lower oil quality [6]. Other oil quality indicator Deterioration of Bleaching Index (DOBI) normally correlated to the ripeness of FFB when harvested [7]. Therefore, harvesting the FFB should be performed when fruits reached optimal ripeness. The ripeness can be determined by the ratio of oil and moisture content in the fruits' mesocarp [7]. When the fruits ripening



oil in the mesocarp increase and moisture content reduced. The reduction of moisture content occurs due to the translocation of oil formation in the fruit [8].

Previously, evaluation of ripeness level, damage, and delaymaned process of FFB had been studied [4, 9, 13], and direct influence to FFB quality and oil extracted were found [4, 7, 10-15]. However, these studies were manually performed, and required high cost and resources. Different methods were developed enable the quality observation of FFB in-situ. The methods delivered the results in rapids and nondestructive nature using NIR spectroscopy [9]. However, the study only performs measurements on FFB at certain temperature. The effect of different temperature regime has not been explored yet.

Reducing fruits temperature immediately after harvest can extend it shelf life [Cecilia, 2008]. Among other methods, freeze storage has the advantages of maintaining the quality of agricultural product [16-17]. The advantages of freeze storage were evaluated on several fruits such as dates and mangoes [16-17]. Dates shelf life when stored at -18 °C can be extended up to 6 months [16]. Sliced mangoes quality can be maintained for 6 months when stored at -30 °C [17]. However, no reports of storing the FFB using freeze method. Therefore, in this research we explore the possibility of storing FFB below the freezing temperature.

For processing frozen foods, acclimatization process was required to restore the product temperature. The temperature set point is unique for each processing procedure. For FFB milling, the temperature should be risen to 121 °C [18]. However, moisture content can influence the temperature profiles of products during acclimatization. When FFB developed, ripening process caused the moisture content in fruits' mesocarp to decrease due to the accumulation of oil. It is shown by changes in physiological properties, such as appearance and chemical compound. Moreover, the different rate of changes during acclimatization process will be difficult to measure the temperature of fruits constantly. Therefore, a reactive, sensitive, and accurate device is needed for observing the changes. With this challenge, we performed thermal imaging measurement to continuously monitor FFB temperature during acclimatization. This technique will be useful for more efficient milling in palm mills.

2. Material and Methods

In this study, Oil Palm FFBs were observed from antheses to harvest. The bunches were selected from 7 years trees or older. The samples located at 0°41'44.6" South and 100°58'54.7" East. The plants were Marihat cultivar and cultivated according to standard practice (regularly fertilized, pruned and irrigated). To determine the effect of ripeness level on the FFB acclimatization the fruits were harvested at two ripeness levels (ripe and unripe).

The unripe FFB harvested between 120 and 140 days after antheses (DAA). In this period, the accumulation of oil begins in the mesocarp and kernel fruit [18]. The accumulation continues until the process of decomposition occurs due to oil degradation in the fruit. The degradation process occurs when the fruits passed its optimum ripeness [19]. Therefore, the second group of samples consist of FFB harvested between 160 and 200 DAA. For each samples group three replications are performed.

Since FFB freshness correlated to its oil quality, all fruits were stored at -20 °C immediately after harvested. This is to prevent degradation of oil in fruits' mesocarp. In this study the FFBs stored for 90 ~ 210 days (Table 1) to maintain the freshness before being milled. For processing the FFB the fruits were acclimatized from storage temperature (-20 °C) to room temperature. A thermal camera was used to monitor changes on fruits temperature continuously. The data was recorded every 4 minutes until room temperature was reached by the fruits. Then, the FFB was sterilized, and temperature observation was performed until the fruits reached 50 °C. Measurements were performed at similar interval.

A simple statistical analysis was done to correlated current fruits temperature and observation from thermal camera. The results were then plotted in to a graph, and the R^2 was determined quadratic polynomial. The samples images from thermal camera then subsequently process using image processing software. The background was segmented and the object features were extracted to obtain R, G and B data. The features then grouped according to FFB ripeness when harvested. The data then correlated to samples temperature, using linear regression model. The FFB thermal properties can be determined based on the coefficient of correlation (R^2) from the model. The R^2 calculated based on quadratic polynomial equation. The model explained the linearity of FFB temperature during

acclimatization from freeze storage until the onset of sterilization. The model can be used in combination with thermal vision to directly identify object temperature under observation.

Table 1. Storage arrangement of FFB according to ripeness and storage-time

Harvested	Days of storage (-20°C)
120 DAA	200
140 DAA	200
160 DAA	210
180 DAA	180
200 DAA	90

3. Results and Discussion

In the beginning the FFB contains high MC and only little oil presented. The MC mostly present in fruits' mesocarp. The fruits start to develop after pollination. Young fruits consist of mesocarp and empty kernel. In this stage, the fruit mesocarp is dominated by fiber and moisture. As fruit ripe the moisture in mesocarp decreases due to the translocation of oil formation [8]. The accumulation of oil in fruit replace the moisture [20], and oil started to developed through synthesis process from 120 DAA [9, 18, 19]. The oil content in fruit changes during the ripening in parallel to the metabolism [19]. This process catalyzed due to physiological and biochemical processes in fruits. The process continuously occurred from antheses until fruit fully ripe [4, 9].

Water's thermal capacity is higher than oil [21]. The difference in thermal properties can be observed using a thermal camera [9]. Thermal cameras record the intensity of infrared radiation emitted by the object. The radiation is captured by the microbolometer sensor made from vanadium oxide []. The signal generated by sensor processed by software and displayed on the device screen. The image display uses a pseudo color technique was used to display the image in color. The image recorded in internal memory and converted into temperature data through linear regression model. Using the model, the thermal camera can precisely measure the sample's surface temperature in a non-contact manner.

In this study, FFB was harvested at different ripeness. Optimal ripeness determined by the ratio of oil and water content in fruits' mesocarp [7]. After harvest, the senescent started and fatty acids immediately form in FFB. Therefore, the bunches should be processed within 22 hours [4]. Formation of fatty acids promote by fruit physiological process, which reduce the quality of oil.

The shelf life of the fruit can be maintained with freeze storage. Freeze storage can extend the life of dates and mangoes by up to 6 months [16-17]. In freeze storage the enzymatic and physiological processes are stopped, so that fruit damage can be prevented. For further processing, acclimatization is needed to avoid damage in fruit cells.

In this study, acclimatization of palm fruits was performed by placing the fruits on petridishes. The dish diameter was 100 mm, and fruits were arranged in single level. The samples then placed inside a sterilizer, where temperature was set to 50 °C. The temperature changes of fruit were measured using a thermometer. At the same time samples were recorded by thermal camera. The observation interval is set every 4 minutes until sample reached 50 °C. The data is presented in Figure 1.

The rate of temperature change in the acclimatization process is closely related to the thermal properties of samples. It is influenced by cells and physical structure, the chemical composition of the object, and MC [22]. When heated, unripe fruits were slower to reach room temperature (25 °C) when compared to the ripe one. Afterward, sterilization process started until samples reached 50 °C. In this stage, the unripe fruits were faster to reach the maximum temperature, while the ripe fruits were slower to get to the same condition. Increase of temperature of the object was faster due to higher MC, with lower oil content. This condition can be found in unripe FFB. Thermal properties have positive correlation with thermal capacity, and moisture absorb heat better at higher temperature than oil [21]. The condition is vice versa when the objects were cooled. In addition the rate of temperature changes influence by FFB ripeness when harvested.

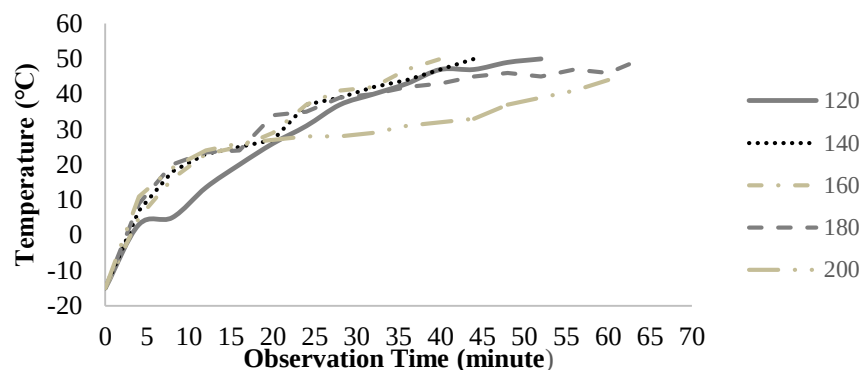


Figure 1. Samples temperature during acclimatization were influence by MC, oil content, and FFB ripeness on harvest

While difference were observed in acclimatization process due to various concentration of oil and moisture in the samples, other phenomena such as hysteresis and nonlinearity can further influence the temperature profile during heating. These two phenomenons generally occur in all cycled process naturally. Chemical composition, physical parameter, and thermal capacities have different nature during heating and cooling process. The difference of sample properties provides challenges to accurately predict the temperature of oil palm during acclimatization process. Continuous measurement is difficult, and therefore, observation of fruit temperature during the acclimatization process requires a device that is reactive, sensitive, and accurate and does not require physical contact.

Thermometer has different sensitivity than thermal cameras. Higher sensitivity is required to rapidly measure temperatures of an object. The thermal camera can respond significantly to the temperature changes in the minute medium under temperature measurements. The sensor used in the thermal camera is a microbolometer made of vanadium oxide with a sensitivity of less than 0.05 °C or equal to smaller than 50 milli K. With this sensitivity the thermal camera can obtained a more linear measurement as compared to regular thermometer, especially when the object is sphere and has small surface area. The comparison of linearity for temperature measurement between thermal camera and thermometer is shown in figure 2.

The level of linearity of the thermal camera in the measurement of the temperature of the palm fruit has a strong attachment ($R^2 \rightarrow 0.9$). Strong attachment indicates that the whole model is able to predict temperature with the help of a thermal camera. The temperature distribution of the object, affected by the composition of the cellulose and lignin ingredients. Cellulose is a fiber that makes up the structure of tissues and strengthens cells [23]. Lignin is part of the cell wall with the most polymers after cellulose [24].

The accumulation of lignin content increases with the development of plant age, while the cellulose content decreases [25]. Lignin is very resistant to degradation, biologically, enzymatically, or chemically [25]. Therefore the rate of change in fruit temperature at the unripe level is faster than fruit at ripeness level. This is caused because the fruit at the unripe level has a low lignin content so that the resistance to heat is low.

The thermal capacity of an object influences the rate of change in the object's temperature. The rate of change in temperature of the palm fruit during acclimatization and sterilization can be seen visually based on the recording results on the visible camera and thermal camera (Table 2). The results of the thermal camera image are then evaluated using the image processing program. Image processing is performed to present the results of thermal images into the color channel values, namely the R, G, and B color channels (figure 3). The value of this color channel is related to the intensity of the radiation captured by the infrared sensor on the camera; the sensor then translates the radiation that hits the sensor surface into an electrical quantity. The electrical scale is then converted into a digital scale with a

grayscale, then this grayscale is converted into a range of standard colors that are in the camera's memory. Of the grayish range is low, the range of colors shown goes to blue, while if the gray level is high, the range of colors shown is towards red.

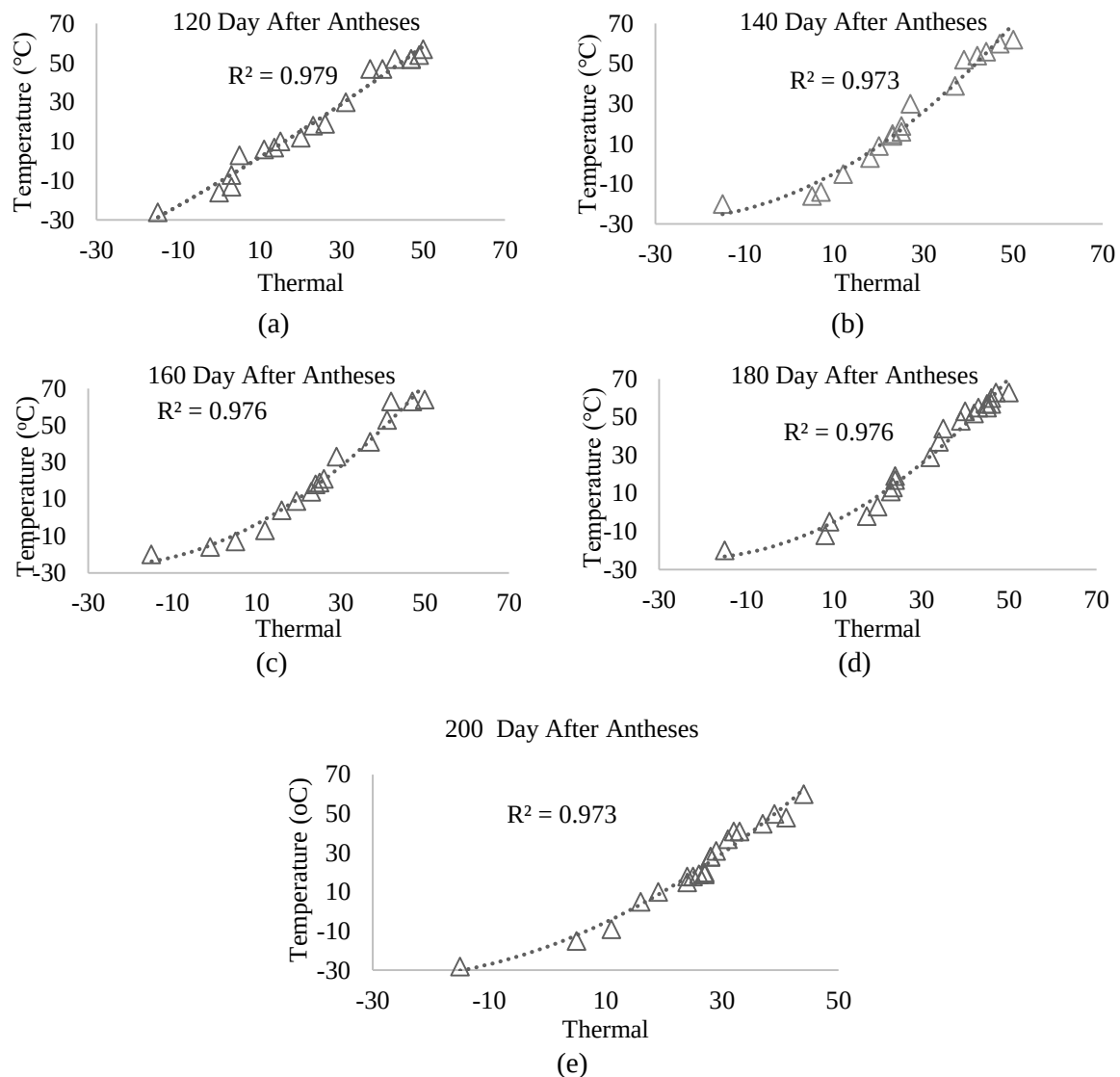


Figure 2. Linearity of The Thermal camera Sensor to the rate of temperature change (FFB) with the ripeness level (a) 120, (b) 140, (c) 160, (d) 180 and (e) 200 days after antheses

The change in color of the palm fruit in the freeze storage is displayed in pseudo color. This display interprets the thermal properties of the material. Thermal properties captured by the camera sensor. Changes in fruit color and temperature are affected by the thermal properties of the material. The thermal properties of the material correlate with the oil and water content. Water has a high thermal capacity than oil. Materials with high thermal capacity will heat faster than low thermal capacity materials. In climatization, fruit sterilization is analogous (continuous), then the difference of concentration is affected by the increase in the rate temperature change in the material.

Color changes correlate with the temperature changes. The value of this color channel is related to the intensity of the radiation captured by the infrared sensor on the camera; the sensor then translates the radiation that hits the sensor surface into an electrical quantity. The electrical scale is then converted into a digital scale with a grayscale, then this grayscale is converted into a range of standard colors that

are in the camera's memory. Of the grayish range is low, the range of colors shown goes to blue, while if the gray level is high, the range of colors shown is towards red. Besides the oil and water content, the material's thermal properties are affected by the quality of the oil. If the quality of palm oil in mesocarp is high, the free fatty acid content is low; if the oil has been degraded or the oil quality has decreased, free fatty acids will be high. If the FFA is high, the heat absorption will increase.

Based on Figure 3, it is estimated that redness, greenish, and bluish palm oil fruit are frozen, then climatized to reach a temperature of 50 °C updated by the ripeness level at harvest. The lower ripeness level, the increased water content. The high level of ripeness, the oil content increases.

But at some age, the process of oil oxidation occurs. This oxidation process is caused by water and oxygen contamination in the air. This contamination is affected by the oil contained in the mesocarp fruit. If the quality of oil increases, the climatization temperature increase process will decrease. This will affect the thermal properties of the fruit.

Based on the linearity of frozen palm fruit, then evaluated against thermal properties captured by the camera in the form of pseudo reddish, greenish, and bluish color with a coefficient (R^2) (180) = 0.8037; (200) = 0.8574; (120) = 0.561. The linearity thermal properties of the fruit are affected by the ripeness level and quality of the fruit.

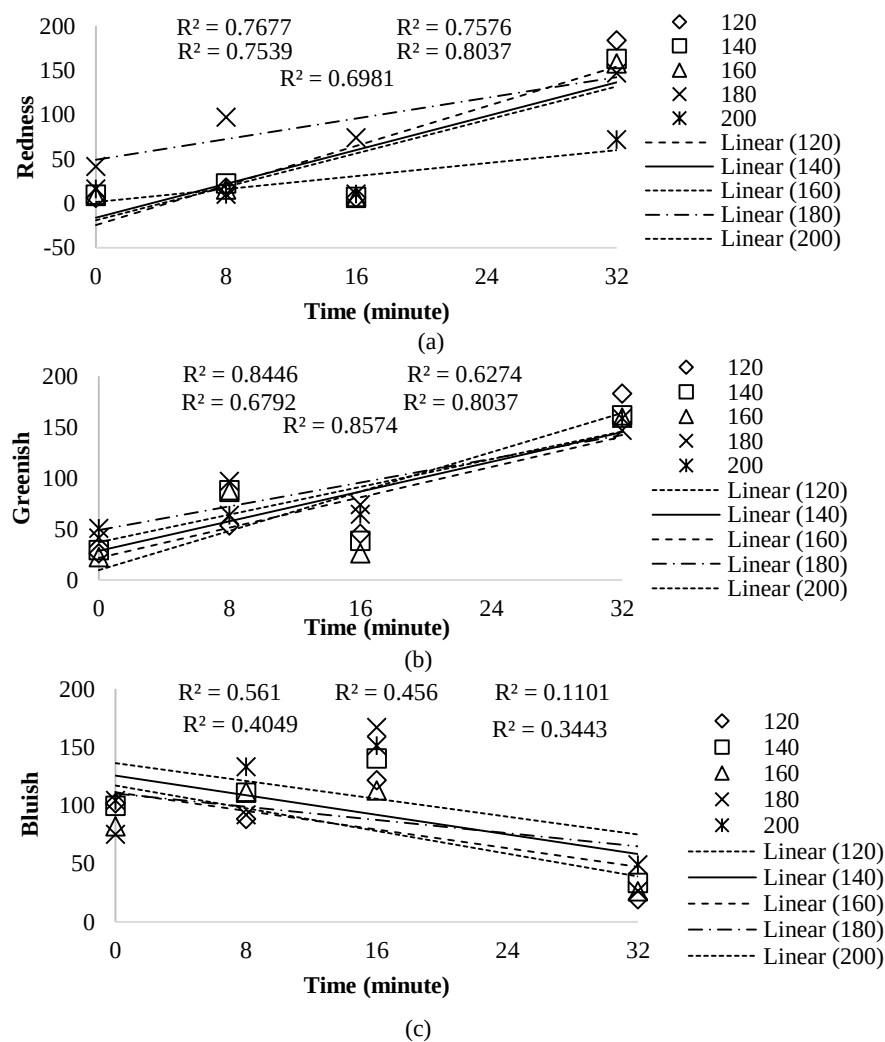
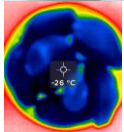
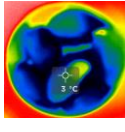
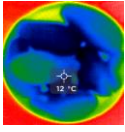
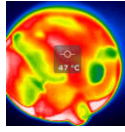
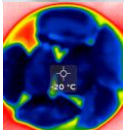
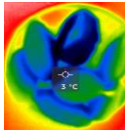
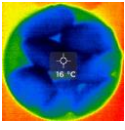
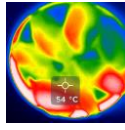
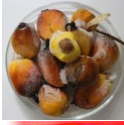
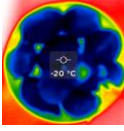
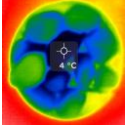
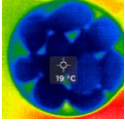
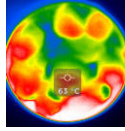
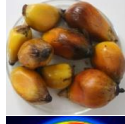
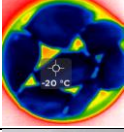
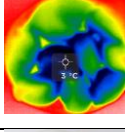
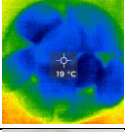
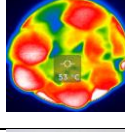
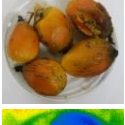
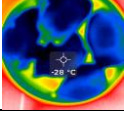
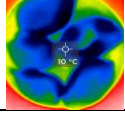
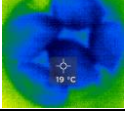
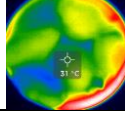


Figure 3. The linearity of palm oil in freeze storage with thermal properties captured by the camera in pseudo colour (a) Redness (b) Greenish (c) Bluish

Table 2. Visualization of the palm fruit image at five of ripeness level using a visible camera and thermal camera

Time		0	8	16	32
Ripeness level/ Type of camera		(-28 ~ -20 °C)	(3 ~ 10 °C)	(12 ~ 19 °C)	(31 ~ 63 °C)
120	Visible camera				
	Thermal camera				
140	Visible camera				
	Thermal camera				
160	Visible camera				
	Thermal camera				
180	Visible camera				
	Thermal camera				
200	Visible camera				
	Thermal camera				

4. Conclusion

In this study, the thermal characteristics of palm fruit were identified based on the rate of change in temperature at five levels of fruit maturity. Thermal properties were observed using a thermal camera and evaluated based on recorded thermal images. The rate of temperature change is influenced by the

chemical composition of the fruit. The chemical composition formed in fruit is related to the development of the ripeness level. Moisture content affects the thermal capacity of the material. Besides, oil content also affects the thermal efficiency of the material. Based on the results of the analysis, the slowest rate of temperature change of palm fruit occurs in the ripe palm. This is affected by the condition of ripe palm, which has a higher lignin content than unripe palm fruit. High lignin content increases fruit resistance to temperature changes. The entire model can predict the temperature of objects using a thermal camera with a level of determination coefficient (R^2) of 0.979, 0.973, 0.976, 0.976, and 0.973 for stored fruits harvested at 120, 140, 160, 180, and 200 days after antheses.

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